

Introduction to Random Forest

Deep Dive into the Intuition behind the Algorithm

CampusX Lecture Series · Supervised Machine Learning

The Core Intuition

Why many weak predictors make one strong model.

| Defining the Random Forest

Ensemble Learning

Random Forest is an ensemble method that combines multiple Decision Trees to perform classification or regression. It operates on the principle of the "Wisdom of the Crowd."

The "Forest" Concept

By building hundreds of independent trees and aggregating their results, the algorithm creates a robust model that generalizes much better than any single decision tree could.

| Three Pillars of Performance



Bootstrapping

Creating diverse training sets by sampling rows from the original data with replacement (Row Sampling).



Random Splits

At every node split, only a random subset of features is considered, ensuring trees are decorrelated.

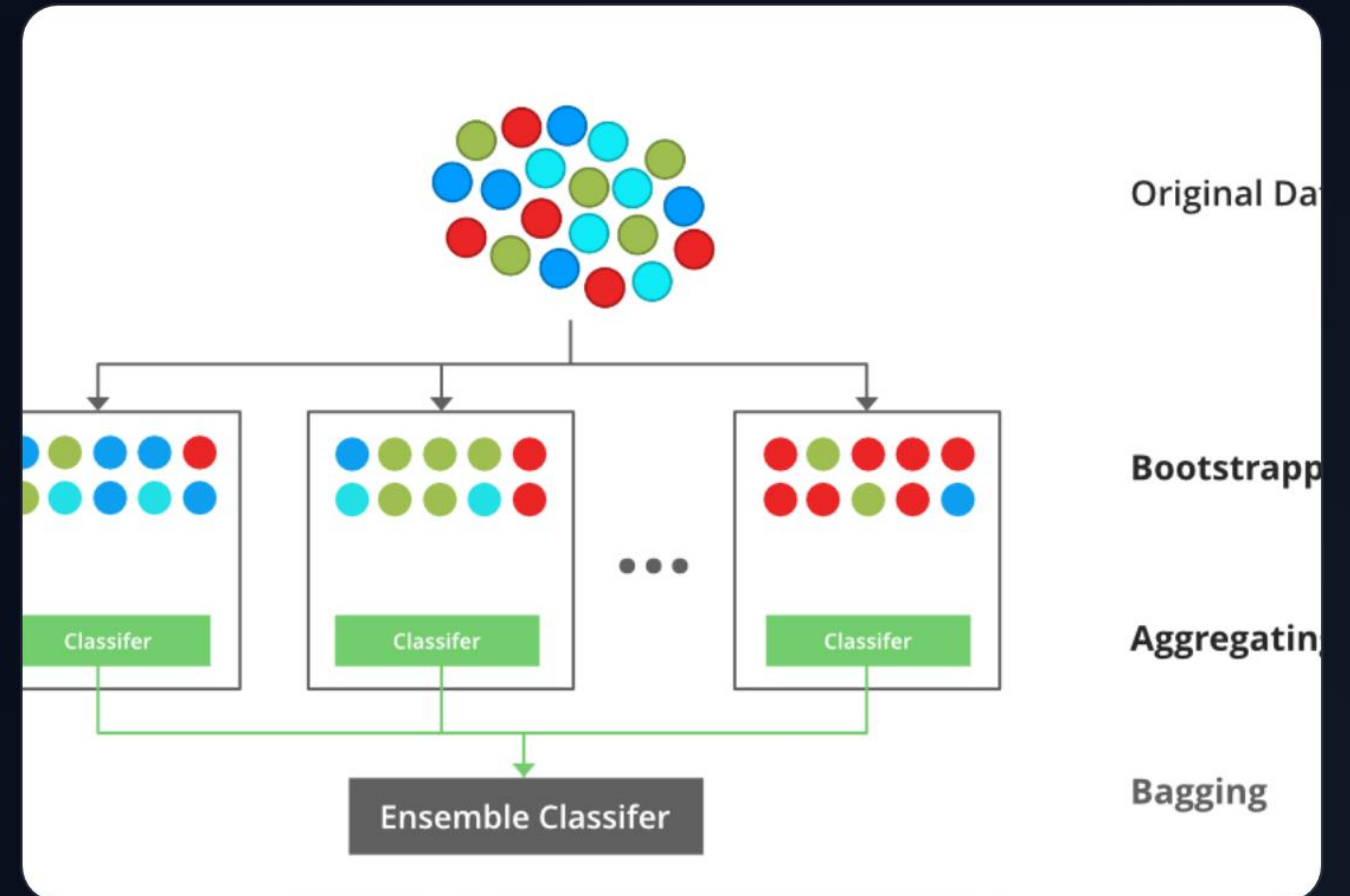


Aggregation

Combining outputs via Majority Voting for classification or Averaging for regression tasks.

| Bootstrap Aggregating (Bagging)

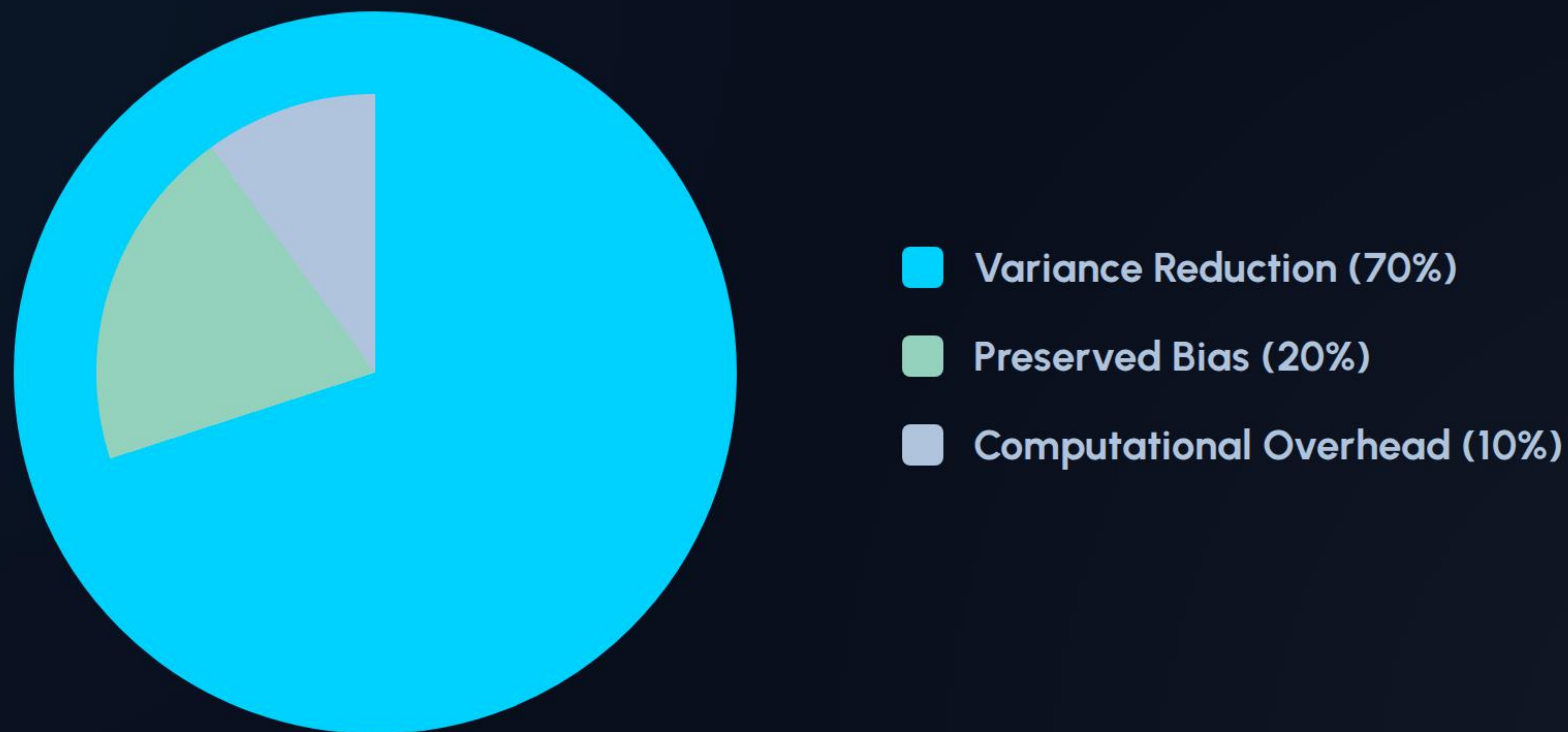
-  **Data Diversity:** Each tree is trained on a unique "Bootstrap Sample."
-  **With Replacement:** Some rows may appear multiple times, while others (OOB) are left out.
-  **Stability:** Bagging significantly reduces the model's sensitivity to outliers in the training data.



| Dual Randomness Strategy

-  **Row Randomness:** Every tree sees a different set of observations, preventing the model from memorizing the noise in a specific subset.
-  **Feature Randomness:** Even at each split, the tree can only choose from a random selection of features. This ensures a dominant feature doesn't mask other useful patterns.
-  **Decorrelation:** The combination of these two ensures that the individual trees are as different from each other as possible.

| The Bias-Variance Trade-off



Random Forest maintains the low bias of Decision Trees while dramatically reducing Variance through averaging.

Model Performance Comparison

Single Decision Tree

72% Accuracy

Bagging (Rows Only)

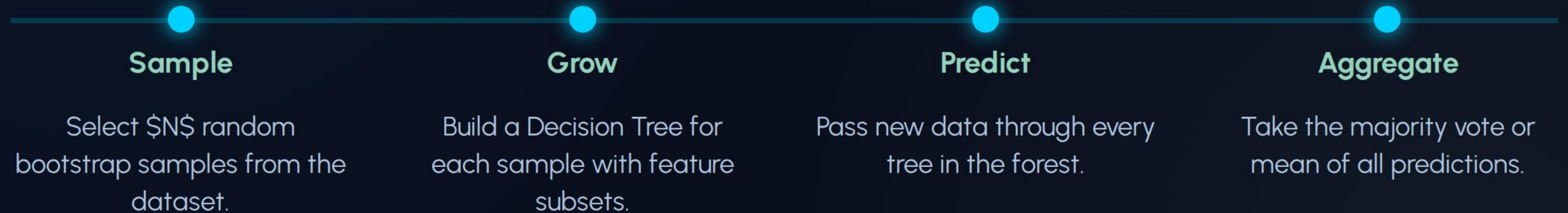
85% Accuracy

Random Forest

94% Accuracy



| Algorithm Step-by-Step



| Wisdom of the Crowd

100+

Individual Estimators

Unmatched Robustness

-  Handles thousands of input variables without deletion.
-  Naturally estimates effective feature importance.
-  Resilient against missing data and noisy outliers.

Key Hyperparameters

Parameter	Description	Impact on Model
n_estimators	Number of trees in the forest	More trees = better performance but slower compute.
max_features	Size of random feature subsets	Controls diversity; usually $\sqrt{M}$ for classification.
max_depth	Maximum depth of each tree	Prevents overfitting by limiting individual tree complexity.
bootstrap	Whether to use bootstrap samples	Enables/disables the core bagging mechanism.

Questions?

Thank you for attending this CampusX lecture!

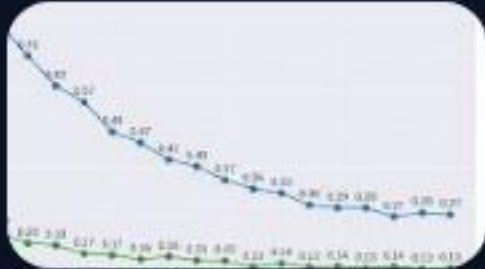


| Image Sources



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Source: medium.com



https://www.qognitiveai.com/blog/tree-benchmarks/DT_RF_features.png

Source: www.qognitiveai.com